ARROWSTACK FULL STACK DEVELOPMENT INTERNSHIP

Project 3 — Subscription Analytics SaaS MVP

Type: Capstone / Portfolio task **Stack:** Node.js + Express REST API · JWT + RBAC · relational store with SQL migrations · React (Vite) SPA with charts · simulated Stripe-like billing

Scenario

A SaaS control plane that manages **plans**, **customer accounts**, **usage records**, **billing events** and **business analytics** (MRR, ARR, ARPA, churn).

Roles

Role	Can do
customer	view own account, record own usage, view own invoices
analyst	read-only analytics and account lists
admin	full control: plans, accounts, usage, billing runs, audit

Core features (per the handbook)

- **Role-based access** — customers scoped to their own account; analysts read-only; admin full.
- **REST API integration** — plans/accounts/usage/billing/analytics endpoints consumed by the SPA.
- **Relational database with migrations** — users, plans, accounts, usage_records, billing_events.
- **Validation & error handling** — e.g. an unknown usage metric -> 400.
- **Audit-friendly activity records** — signups, plan/account/usage/billing actions logged.
- **Deployment-ready configuration** — env-driven port/DB/secret.

The analytics model

- **MRR** = sum of `price_monthly` across **active** accounts.
- **ARR** = MRR × 12.
- **ARPA** = MRR ÷ active accounts.
- **Churn rate** = cancelled accounts ÷ all accounts.
- **Usage analytics** aggregated by metric and by month (period `YYYY-MM`).

Simulated billing run

`POST /api/billing/run` generates one `charge` event per active account for a period: `amount = plan.price_monthly + max(0, usage_units - plan.included_units) × plan.overage_rate`. The run is **idempotent** — re-running the same period creates no duplicates.

Demo logins

Email	Password	Role
admin@arrowstack.in	Admin@123	admin
analyst@arrowstack.in	Analyst@123	analyst
customer@arrowstack.in	Customer@123	customer (Acme Retail)

Run it

```
npm install
npm run start:p3                # http://localhost:4003

cd project-3-subscription-analytics/web
npm install && npm run dev      # http://localhost:5173
```

API summary

Method	Path	Access	Purpose
POST	`/api/auth/register`	public	sign up (optionally creates an account)
POST	`/api/auth/login`	public	obtain JWT
GET	`/api/plans`	any	list plans
POST	`/api/plans`	admin	create plan
GET	`/api/accounts`	admin, analyst	list accounts
POST/PATCH	`/api/accounts` `/api/accounts/:id`	admin	create / change status or plan
GET	`/api/accounts/:id`	any (scoped)	account + usage + invoices
POST	`/api/usage`	any (scoped)	record usage
POST	`/api/billing/run`	admin	run simulated billing (idempotent)
GET	`/api/billing/events`	admin, analyst	billing history
GET	`/api/analytics/overview`	admin, analyst	MRR/ARR/ARPA/churn
GET	`/api/analytics/usage`	admin, analyst	usage by metric & period
GET	`/api/analytics/top-accounts`	admin, analyst	accounts ranked by usage
GET	`/api/audit`	admin	audit trail

Documentation

- Requirements (docs/REQUIREMENTS.md)
- Test evidence (docs/TEST-EVIDENCE.md) — 28/28 checks passing

Source: `server/`, `web/`, `test/run.js`.